Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2001-I-Q05] | Part A (Q1-Q5)

Answer: 937

1. The other vertices are $(u, v), (-u, v)$ with $u > 0$. Equilateral geometry gives $v = 1 - \sqrt{3}u$.

2. Substitution in the ellipse yields $13u^2 - 8\sqrt{3}u = 0$, so $u = 8\sqrt{3}/13$. The side length is $2u$, whose square is $768/169$. Thus the sum is 937.

Source / 来源

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2024-I-Q09] | Part B (Q6-Q10)

Answer: 480

1. Opposite vertices are negatives, and diagonals are perpendicular. Put $B = (x, mx)$; the hyperbola gives $x^2 = 120/(6 - 5m^2)$. The perpendicular diagonal meets the hyperbola only if $5/6 < m^2 < 6/5$.

2. Consequently $BD^2 = 480(1 + m^2)/(6 - 5m^2)$, an increasing function of m^2 on that interval. Its unattained lower limit as $m^2 \to 5/6$ is 480.

3. Every value is larger than 480 and values approach it arbitrarily closely, so the requested number is 480.

Source / 来源

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2025-I-Q09] | Part B (Q6-Q10)

Answer: 062

1. An inverse rotation sends (x, y) to $((x + \sqrt{3}y)/2, (-\sqrt{3}x + y)/2)$. Seek a point whose inverse rotation is its reflection $(-x, y)$ across the symmetry axis; both then lie on the original parabola.

2. This requires $y = -\sqrt{3}x$. With $y = x^2 - 4$, the positive root is $x = (-\sqrt{3} + \sqrt{19})/2$, giving $y = (3 - \sqrt{57})/2 < 0$.

3. The stated uniqueness identifies this fourth-quadrant intersection. Thus $a + b + c = 3 + 57 + 2 = 62$.

Source / 来源

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2026-II-Q06] | Part B (Q6-Q10)

Answer: 050

1. Parameterize the parabola by $(4+u, u^2/2-2)$. At tangency the radius is normal to the parabola, equivalently a critical point of squared distance $D(u) = u^2 + (u^2/2 - 41)^2$.

2. $D\prime(u) = u(u^2 - 80)$, so $u = 0, \pm\sqrt{80}$. Their squared distances are 1681 and 81, giving radii $41, 9$. Their sum is 50.

Source / 来源

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2005-II-Q15] | Part C (Q11-Q15)

Answer: 169

1. The centers are $F_1 = (-5, 12), F_2 = (5, 12)$ and radii $16, 4$. A desired center P has $PF_1 + PF_2 = 20$, giving ellipse $x^2/100 + (y-12)^2/75 = 1$. Every ellipse point is admissible since its PF_1 is between five and fifteen, making circle radius $16 - PF_1$ positive.

2. Substitute $y = ax$: $(3 + 4a^2)x^2 - 96ax + 276 = 0$. Real intersection requires discriminant $4800a^2 - 3312 \geq 0$.

3. The least positive slope is tangent, with $a^2 = 69/100$, so $p + q = 169$.

Source / 来源

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2021-I-Q15] | Part C (Q11-Q15)

Answer: 285

1. Add the first parabola equation to half the second to obtain the circle $x^2+y^2-x/2-41y+400-3k/2=0$. Its squared radius is $325/16+3k/2$, so the radius condition gives $k\le 280$.

2. Let $c=k+20$ and $F(x)=2(x^2-c)^2-x-k$. For $1\le k\le 4$, $F(x)>0$ for $x\le -k$; for $-k<x\le 0$, $x^2<16<c$ gives $F(x)>50-k>0$. Thus there are no negative roots. Its derivative has only one positive zero, so there cannot be four roots.

3. For $k\ge 5$, $F(-\sqrt{c})=\sqrt{c}-k\le 0$, $F(0)>0$, $F(\sqrt{c})<0$, and both tails are positive. For $k>5$, the four sign intervals give four distinct roots. At $k=5$, the root $-\sqrt{c}=-5$ has derivative -1, so the second negative root still exists.

4. Thus the allowed integers run from 5 to 280, and their extreme sum is 285.

Source / 来源

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2022-II-Q12] | Part C (Q11-Q15)

Answer: 023

1. The first ellipse has foci $F=(-4,0), G=(4,0)$ and focal-distance sum $2a$. The second has foci $H=(20,10), I=(20,12)$ and focal-distance sum $2b$.

2. For a common point P, group the four distances as $2(a+b)=(PF+PH)+(PG+PI)\geq FH+GI=26+20=46$. Thus $a+b\geq 23$.

3. Segments FH and GI cross at $P=(14,15/2)$. Its distances to F,G,H,I are $39/2, 25/2, 13/2, 15/2$, so $a=16, b=7$ satisfy the strict parameter restrictions and attain 23.

Source / 来源

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2024-II-Q12] | Part C (Q11-Q15)

Answer: 023

1. Parameterize each segment by intercepts $(\cos\theta, 0), (0, \sin\theta)$, for $0 < \theta < \pi/2$. A positive point (x, y) is on it exactly when $F(\theta) = x\sec\theta + y\csc\theta = 1$. The segment AB corresponds to $\theta = \pi/3$.

2. Both secant and cosecant are strictly convex here, so F is strictly convex and tends to infinity at both endpoints. A point belongs to just one segment precisely when $F(\pi/3) = 1$ is its minimum, requiring $F\prime(\pi/3) = 0$, or $y = 3\sqrt{3}x$.

3. Intersect with AB, whose equation is $y = \sqrt{3}/2 - \sqrt{3}x$: $C = (1/8, 3\sqrt{3}/8)$. Thus $OC^2 = 7/16$ and $p + q = 23$.

Source / 来源